

Project Design Phase-II

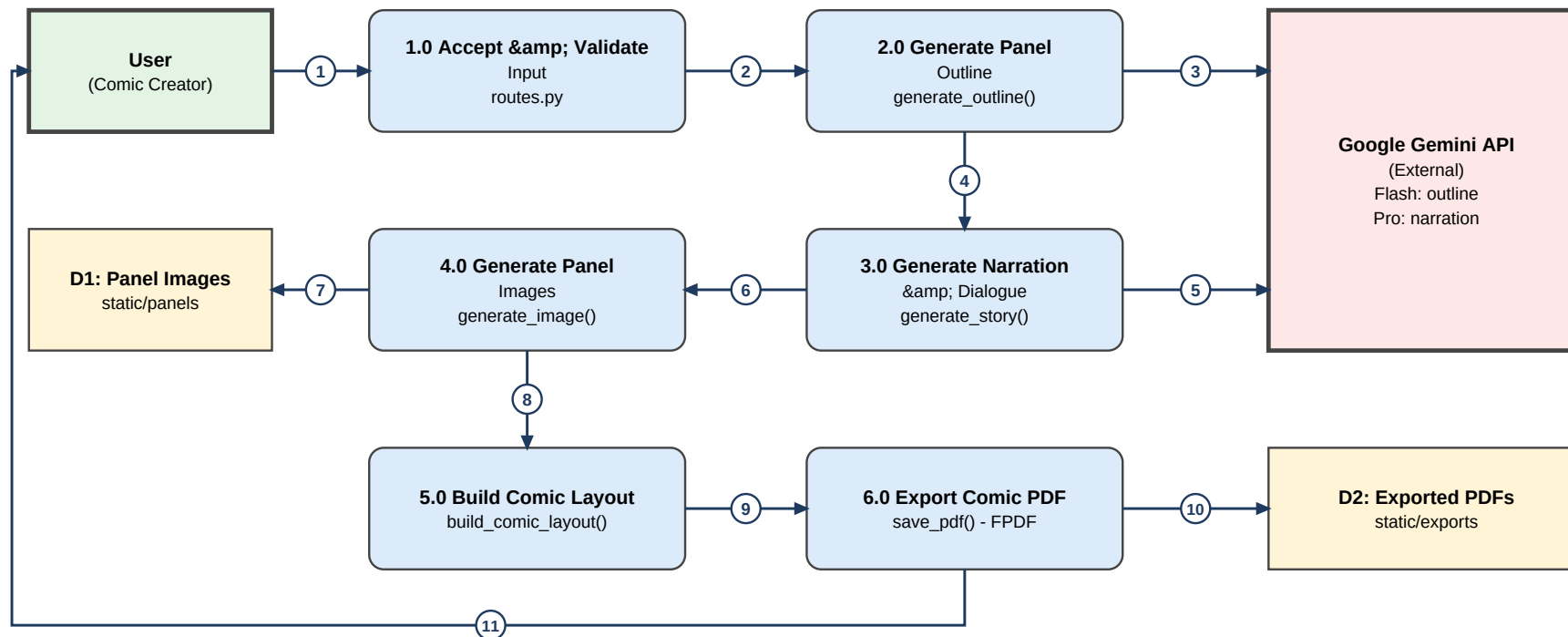
Data Flow Diagram & User Stories

Date	01 October 2026
Team ID	SWTID-2026-8542
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Data Flow Diagrams:

The Data Flow Diagram (DFD) below shows how data enters and leaves ComicCraft, what processes change it, and where it is stored: user story inputs flow through outline, narration and image generation, are combined into a comic layout, and are exported as a PDF.

DFD Level 0 - ComicCraft



- 1 Story prompt, character name, setting, tone, art style (form / JSON)
- 2 Validated user inputs
- 3 Story prompt to Gemini Flash; 5-panel outline returned
- 4 Panel outline (title, scene, image prompt)
- 5 Outline to Gemini Pro; narration and dialogue returned
- 6 Image prompts and story text to image generation

- 7 Generated panel images saved to D1
- 8 Panel images, paths and story text to layout builder
- 9 Structured layout (panel no., image path, text)
- 10 Timestamped PDF saved to D2
- 11 Comic preview, PDF download and export-success page to user

User Stories

Below are the user stories for the ComicCraft product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Comic Creator (Web user)	FR-1 Story Input	USN-1	As a user, I can enter a story prompt, character name, setting, tone and art style in the web form so that my comic is personalized.	The form submits and comic generation starts with my inputs	High	Sprint-1
API Client / Developer	FR-1 Story Input	USN-2	As an API client, I can send my story details as a JSON request to /generate-comic/json.	I receive a JSON response with layout data and the PDF path	Medium	Sprint-2
Comic Creator (Web user)	FR-2 Comic Outline Generation	USN-3	As a user, I get a structured 5-panel outline of my story.	Each panel has a number, title, scene description and image prompt	High	Sprint-1
Comic Creator (Web user)	FR-3 Story Narration & Dialogue	USN-4	As a user, I get narration and character dialogue for every panel.	Text is generated for all 5 panels and follows the outline	High	Sprint-1
Comic Creator (Web user)	FR-4 Comic Illustration Generation	USN-5	As a user, I get a comic-style image for each panel in my chosen art style.	An image is generated and shown for every panel	High	Sprint-1
API Client / Developer	FR-4 Comic Illustration Generation	USN-6	As a developer, I can test image generation with a custom prompt through /test-image.	An image is generated from the prompt without creating a full comic	Low	Sprint-2
Comic Creator (Web user)	FR-5 Comic Preview & Customization	USN-7	As a user, I can preview my comic panel by panel on the web page.	The preview page shows each panel's image with its narration	High	Sprint-1
Comic Creator (Web user)	FR-5 Comic Preview & Customization	USN-8	As a user, I can change the tone or art style and regenerate the comic.	The new comic reflects the newly selected tone and style	Medium	Sprint-2
Comic Creator (Web user)	FR-6 Comic Export	USN-9	As a user, I can download my comic as a PDF with images and narration.	A PDF with a timestamped filename is generated and downloadable	High	Sprint-2
Comic Creator (Web user)	FR-6 Comic Export	USN-10	As a user, I see a confirmation page after exporting my comic.	The export success page is displayed after the PDF is saved	Medium	Sprint-2